Complex Analysis Homework 3

1. Let n be a positive integer.

(a) (i) Find all the roots of $z^{2n}+1=0$.

(ii) By factorizing $z^{2n}+1$ into a product of quadratic factors with real coefficients, prove that

$$z^n+\frac{1}{z^n}=\prod_{k=0}^{n-1}\left(z+\frac{1}{z}-2\cos\frac{(2k+1)\pi}{2n}\right) \text{ for all } z\neq 0 \text{ and}$$

$$\cos n\theta=2^{n-1}\prod_{k=0}^{n-1}\left(\cos\theta-\cos\frac{(2k+1)\pi}{2n}\right)$$

(iii) Using (a)(ii), prove that $\prod_{k=0}^{n-1}\cos\frac{(2k+1)\pi}{4n}=\prod_{k=0}^{n-1}\sin\frac{(2k+1)\pi}{4n}=\frac{\sqrt{2}}{2^n}$.

(b) Let $n > 1$.

(i) Find all the roots of $z^{2n}-1=0$.

(ii) By factorizing $z^{2n}-1$ into a product of quadratic factors with real coefficients, prove that

$$z^n-\frac{1}{z^n}=\left(z-\frac{1}{z}\right)\prod_{k=1}^{n-1}\left(z-2\cos\frac{k}{n}\pi+\frac{1}{z}\right) \text{ for all } z\neq 0 \text{ and}$$

$$\sin n\theta=2^{n-1}\sin\theta\prod_{k=1}^{n-1}\left(\cos\theta-\cos\frac{k\pi}{n}\right)$$

(iii) Using (b)(ii), prove that $\prod_{k=1}^{n-1}\sin\frac{k\pi}{2n}=\prod_{k=1}^{n-1}\cos\frac{k\pi}{2n}=\frac{\sqrt{n}}{2^{n-1}}$.

2. Suppose $f(z)=u(z)+iv(z)$ is an analytic function and the polar coordinate of z is (r, θ).

(a) Using

(i) Method 1: chain rule

(ii) Method 2: the difference quotient $\frac{f(z+\Delta z)-f(z)}{\Delta z}$, as $z+\Delta z\to z=re^{i\theta}$ along the ray $\arg z=\theta$ and along the circle $|z|=r$,

show that

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

(b) Determine the set on which

$$f(z) = \begin{cases} z & \text{if } |z| \le 1 \\ z^2 & \text{if } |z| > 1 \end{cases}$$

is analytic and compute its derivative.

3. Show that the function $f(z) = \sqrt{|\mathrm{Im}(z^2)|}$ satisfies the Cauchy–Riemann equations at $z = 0$, but is not differentiable at $z = 0$.

4. (a) Suppose f is analytic and real-valued in a domain D. Show that f is constant in D.

(b) Suppose f and $\overline{f}$ are analytic in a domain D. Show that f is constant in D.

(c) Suppose f is analytic in a domain D and $|f|$ is constant in D. Show that f is constant in D.

5. Show that the following functions are harmonic, and find its harmonic conjugates:

(a) $xy + 3x^2y - y^3$

(b) $e^{x^2-y^2}\cos(2xy)$

(c) $\tan^{-1}(y/x),\ x > 0$

(d) $\dfrac{x}{x^2 + y^2}$